

LeWAM — Related Work (standalone preview)

1 Related Work

World-action models: video-generative versus latent. A *world-action model* (WAM) jointly models environment dynamics and the action that drives them, $p(\text{future}, \text{action})$, so that a single learned object both predicts what happens next and supplies the control to make it happen. This frames the WAM against two neighbors. Unlike a vision-language-action policy [12, 4], which maps observations directly to actions with no explicit forward model, a WAM carries dynamics and can reason about counterfactual futures. Unlike a pure latent or 3D world model [26, 17], which predicts futures but defers action selection to a separate inverse model or to test-time search, a WAM internalizes the action pathway. The family splits on the medium of the future. *Video-generative* WAMs, exemplified by DreamZero [24], model $p(\text{video}, \text{action})$ with video diffusion, inherit physics priors from internet-scale video, and learn inverse dynamics from their own predicted rollouts so that no test-time optimization is required. *Latent* WAMs, the LeWM [15] lineage we extend, predict in a self-supervised joint-embedding latent rather than in pixels, keeping the model light and the dynamics abstract. Being-H0.7 [14] is a recent large-scale latent WAM, organizing a future-aware latent world-action workspace from 200k hours of egocentric video; like DreamZero it is search-free, discarding its training-only posterior branch and performing no rollout at inference. A common thread thus emerges: the dominant WAMs, whether video-generative (DreamZero) or latent (Being-H0.7), *learn to avoid test-time search*, amortizing planning into a single forward pass. Fast-WAM [25] makes the trend explicit, asking whether world-action models need test-time future imagination at all and answering *no*—uniformly—finding that video co-training rather than test-time rollout drives control. LeWAM is also a latent WAM—a JEPA predictor with a jointly trained action pathway—but it revisits this question for latent models with *explicit* search and answers it *per task family* rather than uniformly: it deliberately retains a short-horizon search loop and asks *when does latent world-model planning actually help, and when should one fall back to a history-conditioned policy?* Where Fast-WAM removes test-time imagination wholesale, LeWAM shows planning still pays off on goal-reaching/geometry tasks and collapses on contact-rich manipulation, and supplies the history-conditioned pathway that carries the latter case. Its contribution is therefore not to be a latent WAM (Being-H0.7 already is one at scale) but to characterize, per task family, where planning over a latent world model pays off and where it collapses—and to supply the history-conditioned pathway that carries the latter case.

JEPA and latent world models for control. A growing line of work learns a world model in a self-supervised joint-embedding (JEPA) latent and plans within it, dispensing with pixel reconstruction. LeWM [15], the latent on which we build, trains a decoder-free predictor for pixel-latent consistency and plans by sampling-based optimization (CEM) against a latent goal-image cost; DINO-WM [26] plans the same way over frozen DINO features with explicitly “no pre-learned inverse models,” and V-JEPA 2-AC [17] extends a video JEPA to a real Franka arm, again with pure CEM at action time. While these systems convincingly show that a JEPA latent supports planning

across navigation, deformables, and continuous control, none carries a learned action prior: the shared pattern is *pretrained features plus sampling-based CEM/MPC with no action model*, which leaves every candidate rollout at the mercy of a dynamics model never trained to model contact. The field supplies its own evidence that this is brittle, V-JEPA 2-AC’s grasp success on a physical arm sitting well below its reaching performance, exactly the regime where rollouts must traverse contact dynamics the latent does not capture. A complementary line improves the latent’s *geometry* for planning rather than its action model: temporal straightening [22] regularizes the JEPA representation so that latent trajectories are locally straight and Euclidean cost approximates geodesic distance, sharpening the goal-image planning landscape and raising goal-reaching success. This is orthogonal to what we find: straightening conditions a geometry in which the task-relevant variable is already present, whereas our robomimic failures are of a different kind that no metric-geometry regularizer addresses—on Lift the success variable (grasp) is *under-exposed* in the single frame, and on Can/Square it is present yet the task is *multi-stage* so a flat goal-image cost cannot decompose it—both recovered instead by our history-conditioned pathway executing the demonstrated sequence. In contrast, LeWAM keeps this JEPA substrate but attaches a jointly trained action pathway and studies, per task family, where the substrate’s planner does and does not work.

Learned policy as a planning prior. A second tradition warm-starts search with a learned policy, the “thinking fast and slow” division of labor introduced by expert iteration [2]. TD-MPC2 [8] and its predecessor [10] learn a Markovian policy $\pi(a | z_t)$ on a decoder-free latent and use it to seed MPPI, and the MuZero family [21] plays the same role for discrete-action MCTS. Recent instances follow suit: policy-guided world-model planning for language-conditioned navigation [5] seeds search from an instantaneous policy, and the JEPA-native Newt [9] pairs a behavior-cloned prior with annealed-constraint planning across a massively multitask, padded-and-masked action space. These priors successfully turn an expensive search into a guided one, but across the cluster the prior is *instantaneous and raw-action*: it conditions on the current latent state and proposes a single next action. The sole precedent that conditions on a past action, MBOP [3] with $a_t = f_b(s_t, a_{t-1})$, is feed-forward in one previous action, non-JEPA, and raw-action; DINO-WM and V-JEPA 2-AC condition their *dynamics* on action history yet carry no prior at all. In contrast, LeWAM’s action pathway is *autoregressive over past actions* $a_{<t}$ on a JEPA latent rather than Markovian, which is what lets it recover the contact state (“am I holding the object”) that a single-frame latent cannot expose. Newt and LeWAM both predict actions on JEPA latents; LeWAM is distinguished by the history and past-action conditioning and by the planning analysis that motivates it.

Amortized and search-free latent action. A third position replaces search outright by directly predicting actions in latent space. The most relevant instance is GC-IDM [18], a goal-conditioned inverse-dynamics model trained on the *same LeWM latent*, which maps (current latent, goal latent, remaining horizon) to the next action and matches or exceeds CEM across navigation, manipulation, and control at roughly two orders of magnitude lower per-decision cost; Diffuser [11] and Decision Diffuser [1] likewise generate trajectories in one shot from return or skill guidance, and DPPO [20] makes the policy itself the planner. These methods compellingly show that amortization can match search at a fraction of the cost, but in making amortization the destination they forfeit any test-time correction when a single forward pass is wrong. In contrast, LeWAM treats amortization as a *warm start to be refined*, with the action pathway proposing and short-horizon search correcting it where a single inverse-dynamics pass is brittle. GC-IDM, sharing our base, is the must-beat amortization baseline against which the warm-start-then-refine wedge has to earn its cost.

Hierarchical, diffusion-execution, and manipulation planners. On contact-rich manipulation specifically, the dominant recipe is to let a world model plan subgoals and a learned policy execute them. WorldDP [7] diagnoses our setting directly, noting that visual world-model MPC is “limited to single-stage tasks such as reaching or grasping” and struggles with multi-stage ones, and its fix is hierarchical: a high-level world model that optimizes feasible subgoals at runtime with a low-level diffusion policy reaching each, over object-centric slot representations. LDP [23] is the closest published latent planner on our benchmark, reporting robomimic image Lift/Can/Square with a diffusion forecaster and diffusion inverse dynamics rather than a JEPA latent plus CEM, while FF-JEPA [16], Diffusion Policy [6], and VQ-BeT [13] round out the execution-side toolkit of expressive, multimodal action models. These planners convincingly solve multi-stage manipulation, but each assumes the failure of flat latent planning rather than mapping it. In contrast, LeWAM proposes no new hierarchy; its contribution on this axis is a per-family analysis of *when flat latent planning fails*. Single-frame goal-image CEM succeeds when the goal encodes a large, visible, data-supported state change (e.g. an OGBench cube carried to a new pose) and collapses to near-zero on robomimic, where the success variable is visually subdominant and the demonstrations are narrow. This both explains why the hierarchical split helps and locates the failure precisely, on a contact-rich arena (robomimic Lift/Can/Square) that the world-model planning literature barely reports.

Representation versus planning. A sharper challenge questions the planning premise itself. MR.Q [19] argues that predictive *representation* learning, not model-based planning, drives scalable multitask continuous control, exhibiting a model-free actor-critic with auxiliary predictive objectives that beats a world-model planner with no planning at all, and traces that gain by ablation to representation quality itself: stripping the predictive objective collapses the latent’s effective rank and idles much of the critic, a shortfall of representational capacity rather than of search. We engage this head-on rather than against it: LeWAM’s action pathway *is* exactly such a predictive representation, and our own multi-seed results are consistent with the challenge, history-conditioned behavior cloning far outperforming online CEM on contact, while the fine A/B choices within the action head (detached versus action-grounded decoder, raw-action versus embedding-target, anti-collapse regularization) move success rate only within training-run noise. Our goal-conditioned head-to-head reaches the same verdict from the opposite side: an amortized, planning-free inverse-dynamics policy on the shared latent matches sampling-based search, and where both flat policies collapse together on contact-rich robomimic the limiter is what the frozen latent exposes about the task, not the choice between Markovian and history conditioning, the same representation bound MR.Q identifies. We therefore make no state-of-the-art success-rate claim. Instead we keep planning where it provably helps under the per-family rule and lean on the history-conditioned action pathway, recovering contact state, where pure search does not. The honest contribution is threefold: the per-family “when does world-model planning help” analysis; the past-action-conditioned action pathway that recovers contact state, where history-conditioned behavior cloning far exceeds CEM; and the reframing of a JEPA world model and its policy head as a single latent world-action model trained jointly, the latent-with-planning-analysis counterpart to a video-generative WAM such as DreamZero [24].

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